LOKLOK Thesis
Supervisor Review Summary

The Social Media Radio Innovation and Its Impact on University Students' Ratings: An Exploratory Study

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This document summarizes the whole thesis for supervisory review. It states the problem, theory, data, methods, principal results, limitations, and the proposition-level verdicts. It is written in English; an Arabic companion is bound after it. All model-based numbers are conditional on the stated design and are not human treatment effects.

Full thesis: 359 pages. Validation: 42/42 checks.

1 Executive summary

The thesis asks whether a social-media radio platform can improve the epistemic quality of what students hear, trust, and learn without destroying engagement. It develops *Epistemic Systems Theory* (EST), specifies the LOKLOK platform, and evaluates the theory in two evidence streams: a measured cross-sectional wave of 52 university students who used Uber in 6th of October City, and a reproducible *calibrated in-silico experiment* built from the field wave, focus-group material, a document corpus, and participant observation.

The measured wave is descriptive. It anchors the level and spread of the model; it does not identify a causal effect. The model comparison is a within-persona 2^3 design: 500 personas, eight A/B/C cells, 18 sessions per cell, eight mathematical filters, a Zewail-inspired attention gate, and a GRM-IRT measurement model. The main result is a large model-based epistemic gain and a measurable engagement cost:

Table 1: Headline model-based results (500 personas, 18 sessions)

Quantity	Estimate	95% interval	Test
$\Delta\text{SR}_{\text{eval}}$	$+0.827$	$[0.779, 0.875]$	$t(499) = 33.93$, $p < 0.001$, $d_z = 1.52$
$\Delta\text{SR}_{\text{comp}}$	$+11.30$	$[10.64, 11.95]$	$t(499) = 33.74$, $p < 0.001$, $d_z = 1.51$
Δ engagement	-0.183	$[-0.188, -0.178]$	$t(499) = -74.41$, $p < 0.001$, $d_z = -3.33$
Bayesian $\Delta\text{SR}_{\text{eval}}$	$+0.8265$	HDI $[0.7793, 0.8764]$	normal summary, not independent evidence

Note. Intervals describe variation across generated personas, not human sampling precision. The engagement result is a cost, not a success criterion by itself.

The central substantive claim is a trade-off, not universal superiority. The full-factorial decomposition shows that the diversity/cooling path (B) carries most of the contrast (main effect $+0.6265$), BRS adds a smaller effect ($+0.1905$), ERIC is near zero alone (-0.0121), and the B×C interaction is $+0.3791$. The strongest driver in the 256-point joint sweep is the diversity-to-credibility weight `cred_dbi` (PRCC $+0.77$). The default contrast sits at the 78.5th percentile of that declared sweep (median $+0.567$, IQR $[0.383, 0.802]$); this is a sensitivity box, not a probability distribution over real platforms.

The thesis is deliberately explicit about what it does not establish. There was no randomized field deployment, no human NMLS validation, no test of measurement invariance in the v8 matrices, and no representative population sample. The semantic/qualitative block is contextual: shuffling the text leaves the contrast near its default ($+0.808$ vs $+0.827$). The 52 respondents set the level and persona spread; the contrast is carried by the mechanism and the declared state-outcome weights. A Gemini persona-agent pilot reproduced the *sign* of the evaluative gain but overstated its magnitude ($+2.650$) and reversed the engagement sign ($+0.163$); it is a single-shot appraisal, not an 18-session simulation and not human data.

2 Problem, questions, and contribution

The thesis starts from a separation in the literature: platform design, audience measurement, and epistemic quality are usually studied in different literatures. Social-media radio combines

algorithmic recommendation, live presentation, audience metrics, and a public-service language. Existing work describes filter bubbles, misinformation diffusion, parasocial relations, and media literacy, but rarely links those mechanisms to a measurable audience rating and a testable platform design.

The research questions are:

1. Does platform architecture shape structural credibility and audience evaluation (EST-P1)?
2. Do diversity and cooling mechanisms constrain epistemic concentration (EST-P2)?
3. Can the proposed instruments meet basic psychometric accountability (EST-P3)?
4. Do the EST constructs integrate into a coherent measurement model (EST-P4)?
5. Which mechanisms and weights drive the model-based contrast?
6. What does a Gemini persona-agent pilot add to the comparison?

The contribution is an integrated framework and a transparent testbed, not a new universal law. It combines: (i) EST and four propositions; (ii) a specified LOKLOK architecture (BRS, MORE, ERIC, Echo Meter, Algorithmic Cooling, Zewail gate); (iii) a measured wave and a qualitative corpus; (iv) a reproducible calibrated in-silico experiment; (v) a GRM-IRT measurement layer; and (vi) explicit claim tiers. The four-stream BGE-M3 embedding is a pipeline feature, not the headline contribution; the source ablation shows that the semantic block is contextual.

3 EST propositions and hypotheses

EST proposes four linked propositions. EST-P1 treats platform architecture as a structural condition of credibility. EST-P2 treats diversity and cooling as constraints on epistemic concentration. EST-P3 treats measurement as an accountability requirement. EST-P4 treats participatory performance as the educational test. The hypotheses are H1 structural credibility, H2 diversity mechanism, H3 psychometric validity, H4 integration, H5 DeGroot conformity (exploratory), H6 scheduling/duration (exploratory), and H7 participatory performance (prospective).

Table 2: Proposition-level verdicts

Proposition	Hypothesis	Evidence	Verdict
EST-P1 architecture shapes credibility	H1, H4, H5	Measured PB–star and BRS associations; model BRS path; no field randomisation	Partial; model-conditional, not a field causal claim
EST-P2 diversity/cooling	H2	MORE/Lagrange is the dominant model path; cooling and echo terms implemented; topology check is network-independent	Supported within the model; field mechanism untested
EST-P3 measurement accountability	H3	High internal consistency; poor single-factor CFA; invariance not estimated	Mixed: reliability supported, structural validity not established
EST-P4 participatory performance	H7	Prospective claim-appraisal task; no human task data	Untested; future study required

Note. H1–H7 are defined in Chapter 1. "Supported within the model" does not mean a human field effect has been estimated.

4 Data and measurement

The measured wave began with 60 raw questionnaires. A 21-field duplicate screen removed eight duplicate vectors, leaving 52 analysis records. The instrument contains an 11-item broadcaster-rating battery (BRS), a five-item perceived-benefit battery (PB), a star rating, ride duration, interaction level, and context items. Internal consistency is high (BRS $\alpha = .957$, PB $\alpha = .947$, $\omega = .964/.962$), but the single-factor CFA fit is poor. The thesis reports reliability and validity separately and does not treat high alpha as structural validation.

The qualitative stream contains a 160-segment FGD transcript, a 185-segment self-report transcript, 60 participant-observation accounts, and 4,828 corpus documents, plus 25 programme briefs. The 52 retained records supply 31 quantitative features; the semantic block supplies 167 features. The total is 198 features.

4.1 Sentence-transformer architecture and process

The semantic encoder is `BAAI/bge-m3`, a multilingual sentence-transformer with an XLM-RoBERTa backbone. It produces a 1,024-dimensional dense vector and supports inputs up to 8,192 tokens. The thesis uses the dense output only; sparse and multi-vector outputs are not used. The encoder is zero-shot: it is not fine-tuned on the thesis data, and no supervised label

enters the embedding step.

The process is: assemble 5,258 document texts; clean and truncate each text to the model limit; tokenise and encode mini-batches on a Colab T4 in mixed precision; pool and L2-normalise the dense representation; cache vectors by a hash of model name and document set; project the semantic block to 32 principal components; and combine the 167 semantic features with the 31 quantitative features. The cache makes the run reproducible without a second API call. This is a measurement pipeline, not a claim that the 1,024 dimensions are psychological constructs.

4.2 Embedding and latent diagnostics

The 5,258 vectors are already unit length. The first two principal components explain 44.9% of the variance (36.9% and 8.0%). Within-stream cosine similarity is highest for participant observation (0.785), followed by stimuli (0.568), corpora (0.559), FGD (0.531), and self-report (0.466). Cross-stream similarity is lower (0.38–0.48), and the cosine silhouette is 0.196: moderate, not clean, separation. Figure 1 shows the vector space and the similarity matrix.

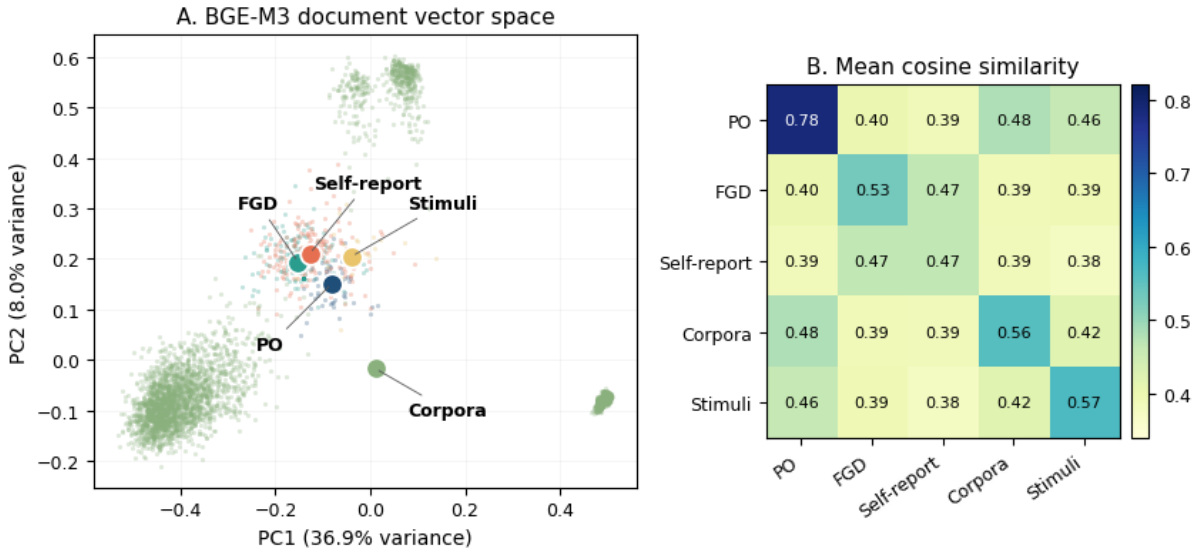


Figure 1: BGE-M3 document vector space (A) and mean cosine similarity (B). The projection is descriptive, not a validated psychological map.

The six-axis manifold is a debiasing coordinate system, not a six-factor psychological model. For the 52 anchors, the covariance condition number is 4.815 and the largest absolute off-diagonal correlation is 0.434. The first two principal components explain 49.9% of the latent variance. Mean within-university distance is 3.211 and mean between-university distance is 3.295 (ratio 0.974): university labels do not separate the anchors. The 500 personas are draws from the fitted Q_0; the figure shows the 52 anchors, not 500 respondents.

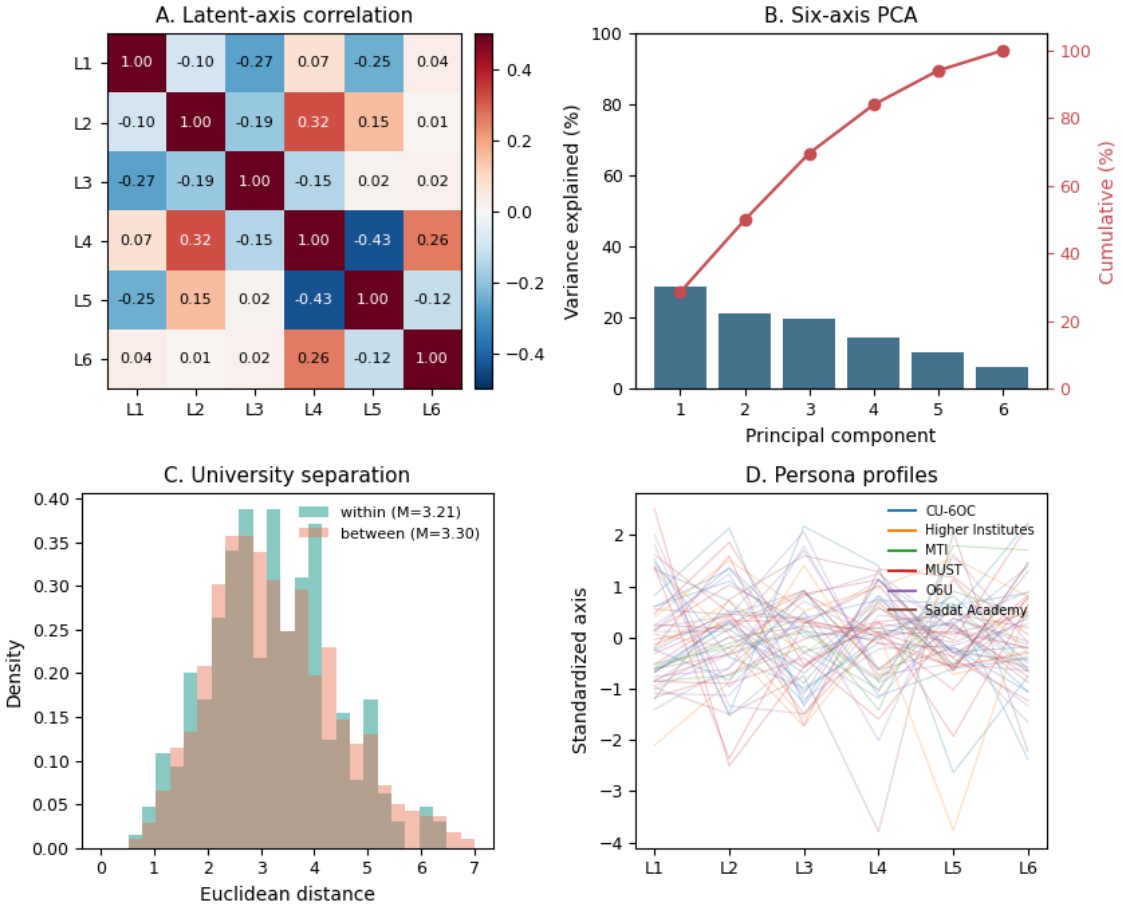


Figure 2: Latent-space diagnostics for the 52 anchors: axis correlations (A), PCA variance (B), within/between university distances (C), and standardised profiles (D).

5 Model and inference

The pipeline applies DML and IPW debiasing, then a spectral manifold (PCA + LLE + Laplacian eigenmaps). The maximum nuisance correlation falls from 0.887 to 0.042 after DML; IPW reintroduces some imbalance (0.127), which is reported rather than hidden. The retained six-axis condition number is 4.82. A Gaussian KDE (or a shrunken multivariate normal if ill-conditioned) defines Q_0; 500 personas are drawn.

Each persona is cloned across the $2^3 = 8$ A/B/C cells and run for 18 sessions. The state tensor is propagated through eight filters: Hamiltonian, Laplace, Lyapunov, DeGroot, Lagrange, Hilbert, Percolation, and Entropy. The Zewail pump–probe gate multiplies the evidence anchor and attentive-retention term. The terminal state is scored by a Samejima GRM-IRT model for the 35-item NMLS and the 8-item MCS+PB battery. The G0 level is anchored to the measured PB mean of 5.119; the treatment weights are declared design inputs, not estimates from the 52 respondents.

Inference is reported in two frameworks. The frequentist analysis uses paired t-tests, Cohen's d_z, 95% repeated-sampling intervals, a full-factorial decomposition, and Wasserstein distances. The Bayesian analysis is a one-sample normal summary of the same paired differences. It is a model-internal re-expression, not independent evidence; its interval width depends on the chosen

persona count. The joint sweep, not $P(\Delta > 0)$, is the relevant uncertainty statement.

A Gemini persona-agent pilot used 24 measured personas × 8 conditions = 192 calls. It is a single-shot appraisal of the condition description, not an 18-session LLM simulation. It tests whether the direction of the result survives a different agent formalism.

6 Results

6.1 Main contrast and decomposition

The main results are in Table 1. The evaluative gain is +0.827 points on the 1–7 scale, the competence-format gain is +11.30 points on the 0–100 scale, and engagement falls by −0.183 on the 0–1 scale. The engagement cost is substantial in standardized terms ($d_z = -3.33$).

The factorial decomposition is: A +0.1905, B +0.6265, C −0.0121; A×B +0.0216, A×C −0.0084, B×C +0.3791, A×B×C +0.0886. MORE dominates; BRS adds a smaller main effect; ERIC matters conditionally through diversity and cooling. Because B×C is nearly half of the total contrast, a share-of-main-effects attribution is misleading and is not used.

6.2 What drives the result

The leave-one-filter-out ablation shows that only DeGroot, Lagrange, and Hilbert change the terminal contrast in this wiring. Disabling DeGroot lowers $\Delta\text{SR}_{\text{eval}}$ by 0.233 [0.197, 0.269], disabling Lagrange by 0.750 [0.713, 0.787], and disabling Hilbert by 0.453 [0.402, 0.503]. Lagrange is the operationalisation of factor B, so the Lagrange-off run is a second switch-off of the diversity objective, not independent evidence for a separate filter. Laplace, Percolation, and Entropy leave the contrast unchanged, and Lyapunov is negligible. The Hamiltonian is a diagnostic and does not enter the outcome.

The source ablation shows that the quantitative block carries most of the contrast. Dropping all semantic features changes ΔSR by only -0.009 $[-0.029, 0.010]$; shuffling the semantic matrix changes it by -0.019 $[-0.039, 0.001]$. The shuffled-text and quantitative-only runs remain close to the full contrast (+0.808 and +0.818). The 52 respondents therefore set the intercept and the persona spread; the contrast is carried by the mechanism and the state-outcome weights. The semantic block is contextual, not load-bearing.

The 256-point joint sweep varies six mechanism parameters and five state-outcome weights, including zero. The contrast ranges from −0.244 to +1.810 (mean +0.593, median +0.567, IQR $[0.383, 0.802]$); 96.5% of draws are positive. The default +0.827 sits at the 78.5th percentile. The top PRCCs are `cred_dbi` +0.77, `cred_brs` +0.31, and `cred_echo` −0.27. The negative echo value is a sign convention: the weight itself is negative. In one sentence: a diversity-seeking selection rule raises chosen diversity, and chosen diversity is assumed to raise perceived credibility. That is the causal chain the thesis tests, not a claim that the field data identified the weight.

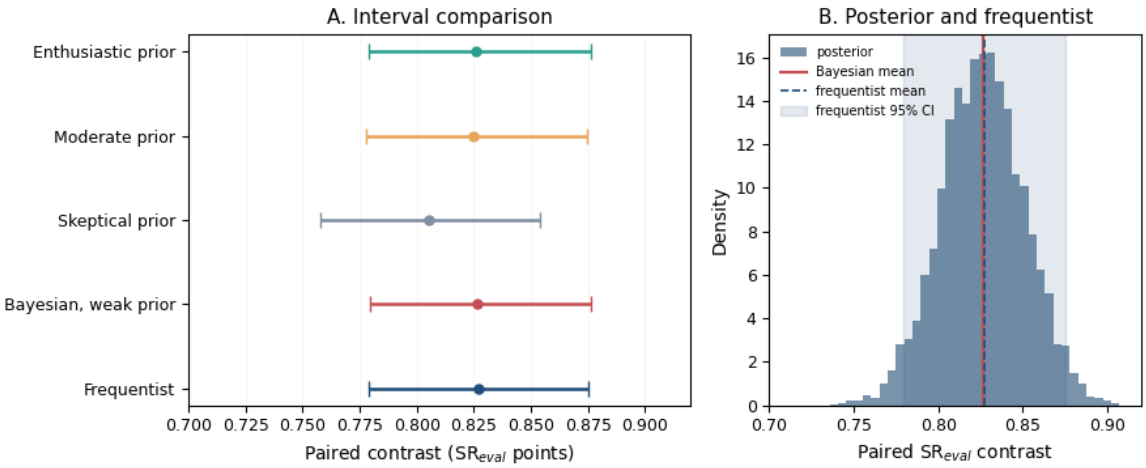


Figure 3: Frequentist and Bayesian summaries of the primary paired contrast. The intervals nearly coincide because they use the same simulated differences; this is not independent evidence.

6.3 Robustness and the semantic question

The contrast is stable to the 60-row file (+0.808 [0.760, 0.856] vs +0.827), to 50 anchor-bootstrap resamples (mean +0.839, 95% [0.805, 0.884], 100% positive), and to ER/WS/BA topology (+0.827/ + 0.826/ + 0.832). The topology result follows from structure: the graph enters only through BRS diffusion, and A is the smallest effect. The dominant B path is network-independent; the topology check confirms that structure rather than adding an independent robustness result.

The held-out predictive comparison uses 52 rows and 198 features. Psychometrics alone predict PB best ($R^2 = .915$); adding 167 semantic features lowers PB R^2 to .583, and semantic-only has negative R^2. In this sample, the semantic block does not improve held-out prediction. With 52 rows and 167 features, that is expected from overfitting alone; it is not evidence that semantics are uninformative in a larger sample. The content-supply stress test sets a heuristic poor-content share of 0.763 from the Uber-review star mean of 1.69; it is an arbitrary stress parameter, not a measured radio-propaganda rate. It lowers ΔSR to +0.636, paired −0.191 [−0.261, −0.122]; the sign remains positive.

6.4 Gemini pilot

The Gemini arm reproduced the direction of the evaluative gain but overstated it (+2.650 [2.342, 2.967] vs +0.827), overstated the competence gain (+31.15 vs +11.30), and reversed the engagement sign (+0.163 vs −0.183). The eight-cell SR_{eval} comparison had MAE 0.655, RMSE 0.842, Pearson $r = 0.269$, and Spearman $\rho = 0.071$. A single-shot LLM appraisal sees the explicit architecture description but does not run the state updates, echo accumulation, or attention gate; it has no mechanism for the engagement cost. The pilot is a methodological triangulation, not human data and not an 18-session simulation.

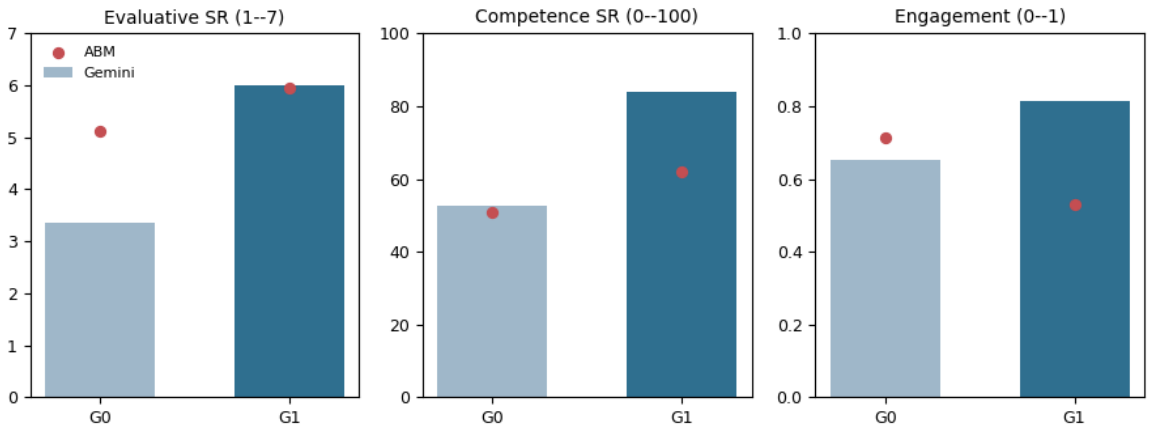


Figure 4: Gemini persona-agent pilot versus the rule-based ABM. The LLM reproduces the direction of the epistemic gain but not its magnitude or the engagement trade-off.

7 Limitations and what remains untested

The thesis does not estimate a human causal effect. The 52-student wave is cross-sectional, convenience-based, and non-randomised. The 500 personas are a computational population, not 500 respondents. The synthetic NMLS profile is a measurement diagnostic, not human media literacy. The Hamiltonian is a formal diagnostic, not a physical law; its sign did not decrease. The BRS is a proposed metric, not a validated consequential ranking instrument. Multigroup invariance is not estimated for the v8 state-mediated matrices. The semantic block is contextual, and the content stress test is arbitrary.

The model's narrow intervals describe simulation variation; they are not human sampling intervals. The Bayesian summary is not independent evidence. The model weights are declared assumptions. The field cannot yet identify the dephasing time, the true poor-content share, or the effect of a real LOKLOK deployment. Those are future empirical questions.

8 Conclusion and defense readiness

The defensible conclusion is that architecture, audience judgment, and measurement should be studied together. EST supplies a falsifiable language; LOKLOK supplies a specified design; the measured wave supplies anchors and limits; the calibrated in-silico experiment shows what the specified design implies. The central finding is the epistemic–engagement trade-off, not a claim that architecture determines human outcomes. The strongest EST propositions remain conditional, and the human causal claim remains future work.

For the viva, the three numbers to remember are $+0.827$ evaluative rating points, -0.183 engagement, and 96.5% positive joint-sweep draws. The four questions to expect are: (i) Lagrange-off removes most of the effect, so is this just MORE? Yes, and that is the design claim: diversity-seeking selection plus a credibility weight. (ii) Shuffling the text gives the same answer, so what did the data do? They set the level and persona spread, not the contrast. (iii) Why do the default weights sit at the 78th percentile of the sweep? They are declared design values, and the sweep brackets them. (iv) Why is the baseline gate open only about 15% of the time? It is a

strong attentional filter; the dephasing time is an assumption, not an empirical estimate.